# U.S. ARMY RESEARCH LABORATORY (ARL / ARO) BAA (ARL-ARO-BAA-2026-AI): TECHNICAL WHITE PAPER
**Title:** Project Ebony v8.0.0: Byzantine Fault-Tolerant Edge Consensus, Unidirectional Cognitive Isolation & Sub-Millisecond SHA-256 Merkle DAG Provenance
**Submitting Entity:** Humphrey Virtual Farms LLC (‘CAGE: 1AHA8‘ | ‘SAM.gov UEI: S1M4ENLHTDH5‘ | Madill, OK 73446)
**Principal Investigator:** Jeffery Humphrey, CEO | [REDACTED-FOR-PUBLIC-RELEASE]
**Research Category:** Defense Basic & Applied Research (‘6.1 / 6.2‘ | Current Readiness: ‘TRL 4‘ -> Target: ‘TRL 5/6‘)
**Cross-Referenced Filings:** DoD CDAO Tradewinds Docket ‘10-26-4014‘ (‘Status: Received‘) & USACE ERDC Rolling BAA (‘2026‘)
**Shared Core C2 Merkle Root (SHA-256):** ‘bb39eeaa66a39ad2e0c0ab74ba4a5c4faac2bf0d98babc0362482b97607151a1‘
**Proposed Phase-1 Effort:** ‘12 Months‘ | ‘$750,000.00‘ Fixed-Price (‘$150,000.00‘ Day 15?30 Mobilization | ‘0%‘ Cost Share | ‘DFARS 252.227-7018‘)

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## 1. Scientific Objective & Contested Tactical Edge Challenge
Tactical Army C2 and autonomous robotic platforms operating under severe electromagnetic emission control (‘EMCON‘) and Denied, Degraded, Intermittent, and Limited (‘DDIL‘) links cannot rely on ‘>250ms‘ commercial cloud AI wrappers or unverified probabilistic LLM outputs to govern physical actuators. When adversary electronic warfare or cyber injection corrupts edge sensor streams, conventional neural controllers lack deterministic mathematical bounds to prevent unsafe physical actuation.

## 2. Mathematical & Architectural Innovation (TRL 4 Verified)
Project Ebony v8.0.0 establishes a provably isolated Three-Brain cyber-physical architecture verified across ‘636‘ compiled Python C2 modules, ‘85 / 85‘ automated unit test suites, and ‘1,420 / 1,420‘ verified engineering assessment items:
* **Unidirectional Optical/Memory Cognitive Isolation (Brain Two -> Brain One):** Brain Two (Analyst) executes offline multi-spectral drift and structural anomaly inference over a strictly read-only memory bus. It possesses zero physical execution path to tactical actuators; all advisory flags must satisfy Brain One’s deterministic physical safety laws (‘<50us‘ loop; modeled ‘36.75ns‘ SiC MOSFET cutoff).
* **Sub-Millisecond Byzantine WORM Merkle DAG Chaining (Brain Three ? Arbiter):** Every state vector from Brain One and Brain Two is cryptographically chained into a 52-table bare-metal SQLite Write-Once-Read-Many (WORM) Merkle Directed Acyclic Graph (‘<0.2ms‘ latency) bound to ‘127.0.0.1‘. Single-byte tampering triggers immediate memory zeroization (‘17 / 17‘ Red-Team vectors blocked).

## 3. Phase-1 Research Schedule & Budget ($750,000.00 / 12 Months)
* **Q1 ($200,000):** Formal mathematical verification of unidirectional memory isolation and Byzantine consensus bounds under simulated DDIL packet loss.
* **Q2 ($200,000):** Hardware-in-the-loop (HIL) benchmarking of ‘<0.2ms‘ SHA-256 WORM Merkle DAG chaining on low-SWaP (‘28.5W‘, ‘24V DC‘) tactical edge processors.
* **Q3 ($175,000):** Adversarial Red-Team fault-injection and single-letter tamper zeroization stress testing across multi-node tactical clusters.

* **Q4 ($175,000):** TRL 5/6 laboratory demonstration and ARL technical monograph delivery.